

A Dual-Branch Interactive Fusion Network to Remove Artifacts From Single-Channel EEG

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Abstract—Electroencephalogram (EEG) obtained from the scalp is often polluted with various artifacts. Deep learning (DL) has been preliminarily explored in EEG denoising and proved to be more promising than traditional methods. However, existing DL methods focus on recovering the clean EEG, while ignoring the recovery of artifacts present in contaminated signals. This limits their ability to eliminate artifacts. In this article, we propose a dual-branch interactive fusion neural network, called EEGIFNet, to remove artifacts from the single-channel EEG. In EEGIFNet, the two branches are employed to recover the clean EEG and residual artifacts, respectively. Convolution layers and bidirectional gated recurrent unit (BiGRU) are cascaded in each branch to capture local and global features. We also introduce an interaction module to extract residual components from opposite branches, which benefits both branches from each other. In addition, a fusion module is designed to adaptively fuse the outputs of the two branches and finally reconstruct the clean EEG. We evaluate the proposed method on publicly available datasets including the electrooculogram (EOG), electromyography (EMG), electrocardiogram (ECG), power line interference (PLI), and motion artifacts. Extensive experiments indicate that EEGIFNet outperforms the state-of-the-art methods on diverse artifacts and signal-to-noise (SNR) levels.

Index Terms—Artifact removal, dual-branch, electroencephalogram (EEG), fusion, interaction.

I. INTRODUCTION

ELECTROENCEPHALOGRAPH (EEG), as a common measurement modality for recording the electrical activity along the scalp, has been widely applied in cognitive science research, brain-computer interface, sleep monitoring, affective computing, disease diagnosis, and so on. In the meantime, new

EEG acquisition instruments have been developed to measure high-quality EEG signals even in mobile wearable situations. Nevertheless, with weak amplitudes, EEG is easily polluted by various artifacts, such as the electrooculogram (EOG), electromyography (EMG), electrocardiogram (ECG), power line interference (PLI), and motion artifacts [1]. The existence of these artifacts will mislead subsequent EEG analysis, limiting the practical application of the EEG. Thus, signal denoising, as one of the important topics in the field of instrument and measurement (I&M), is crucial for EEG signal processing.

Traditionally, similar to denoising other types of signals in the field of I&M, blind source separation (BSS) methods are widely used to remove artifacts from EEG, such as independent component analysis (ICA), canonical correlation analysis (CCA), and independent vector analysis (IVA) [2], [3], [4]. These methods decompose the noisy EEG into a set of source components. The components associated with the artifacts are identified and rejected. BSS requires the number of channels to be more than or equal to that of sources, which is demanding and not suitable for portable scenes with only single or fewer EEG channels [5]. The hybrid methods consisting of signal decomposition and BSS can effectively break through this limitation, such as the combination of the wavelet transform (WT) with ICA, ensemble empirical mode decomposition (EEMD) with CCA, EEMD with IVA, and singular spectrum analysis (SSA) with ICA [6], [7], [8], [9]. These methods decompose the single-channel EEG into multichannel signals, which are further fed into the BSS methods mentioned above. However, the choice of these methods depends heavily on the type of artifacts. For instance, ICA-based methods are recommended to remove EOG artifacts, while CCA and IVA are more suitable for removing EMG artifacts. Almost no method can be applied to multiple types of artifacts without manual adjustment.

Recently, deep learning (DL) has been preliminarily explored in EEG denoising and proved to be more promising than traditional methods. Yang et al. [10] employed a fully connected neural network (FCNN) to eliminate EOG artifacts automatically. Zhang et al. [11] compared several benchmark networks for removing EMG and EOG artifacts, including the FCNN, convolution neural network (CNN), and recurrent neural network (RNN). Sun et al. [12] proposed a multiscale residual CNN to automatically remove the ECG, EMG, and EOG from the contaminated EEG. Sawangjai et al. [13] utilized a generative adversarial network to eliminate

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EOG artifacts, forcing the generated EEG to approximate the real EEG. Zhang et al. [14] defined artifact removal as a two-stage problem involving identification and removal and used GRU with an attentional mechanism to eliminate five kinds of artifacts. Existing DL methods have shown great improvements over traditional methods. However, they focus on recovering the clean EEG from contaminated signals, while ignoring the reconstruction of residual artifacts. For example, EOG and motion artifacts are low-frequency signals with simple waveforms. ECG and PLI have obvious periodicity. Compared with them, the EEG is more complex and irregular. It may be easier to recover them from contaminated signals than a clean EEG.

In the field of image denoising and speech enhancement, DL is often used to directly predict noise and obtain clean signals by subtracting predicted noise from original data. Zhang et al. [15] proposed denoising convolutional neural networks (DnCNNs) to remove Gaussian noise from images, which introduces residual learning to implicitly remove the latent clean image during feature extraction and further predict noise residuals. Odelowo and Anderson [16] first used a noise prediction model with a time-domain subtraction strategy for speech enhancement. Zheng et al. [17] designed two subnetworks to reconstruct the speech spectrum and noise spectrum, respectively, and then recovered clean speech waveforms based on them. The rationale behind the noise prediction methods is that learning the mapping from noisy signal to the features of added noise may be easier than to the features of clean signal [16]. At present, there is no study to reconstruct the clean EEG from the perspective of predicting artifacts.

Inspired by the above, in this article, we propose a dual-branch interactive fusion neural network, called EEGIFNet, to remove artifacts from the single-channel EEG. In EEGIFNet, the two branches are employed to recover the clean EEG and residual artifacts, respectively. Convolution layers and bidirectional gated recurrent unit (BiGRU) are cascaded in each branch to capture local and global features. Besides, since the EEG branch and artifact branch share the same input, the extracted features from them have a residual correlation. There are redundant artifact components and missing EEG components in the EEG branch inevitably, and vice versa. We designed the interaction module to introduce the information exchange between the two branches. Such an interaction can help recover the remaining components from each other. After that, we can obtain the predicted EEG in two ways. One is directly generated by the EEG branch, and the other is obtained by subtracting the generated artifact from the contaminated signal. Lastly, we add a fusion module that adaptively combines the two outputs to reconstruct the final clean EEG. The main contributions of our work can be summarized as follows.

- 1) We introduce a different route for DL-based EEG denoising by recovering residual artifacts from original signals and developing a dual-branch interactive fusion neural network to recover clean EEG and residual artifacts simultaneously. It fuses the outputs of two branches to improve the performance of artifact removal.

- 2) We evaluate the proposed model on publicly available datasets including the EOG, EMG, ECG, PLI, and motion artifacts. The results demonstrate our model achieves state-of-the-art performance on diverse artifacts.
- 3) We also investigate how to choose between EEG reconstruction and artifact reconstruction for different artifacts and signal-to-noise ratio (SNR) levels, which can serve as a reference for future DL work on EEG denoising.

The remainder of this article is arranged as follows. We introduce the proposed EEGIFNet model in Section II. Then data collection and experimental setup are described in Section III. In Section IV, we analyze and discuss the experimental results. Finally, the article is summarized in Section V.

II. METHODS

The overall framework of EEGIFNet is shown in Fig. 1. It mainly consists of the EEG branch, artifact branch, interaction module, and fusion module. The two branches predict EEG and artifact, respectively. The interaction module provides information exchange and feature enhancement for the two branches. The fusion module is used to combine the two outputs and obtain the final pure EEG. We will introduce each part in detail.

A. EEG Branch and Artifact Branch

Similar to the existing DL methods, the EEG branch maps the raw signal to clean EEG. To make full use of the characteristics of both the EEG and artifact, we added another branch to learn the mapping from the raw signal to the residual artifact. In our work, the two branches have the same network structure but do not share parameters.

As we all know, the CNN is good at extracting local features due to its local receptive field, while the RNN is suitable for extracting global features due to the gate structure. To capture the details and contours of the EEG and artifact, we design cascaded convolutional RNNs for each branch, which contains three convolutional layers, one BiGRU layer, and three fully connected layers.

Suppose that $y \in \mathbb{R}^{1 \times T}$ denotes the input contaminated EEG, where T represents the number of sampling points. Taking the EEG branch as an example, local features $F_E = [f_1, f_2, \dots, f_d] \in \mathbb{R}^{C \times d}$ can be obtained after passing through three convolutional layers, where C is the number of channels, and d is the length of the feature. f_t ($t = 1, 2, \dots, d$) represents the extracted feature within a local segment. The length of the segment is determined by the size of the convolution kernel. To capture global information, the BiGRU takes the current input f_t , the forward activation $\overrightarrow{h}_{t-1}$ at the previous moment, and the backward activation $\overleftarrow{h}_{t+1}$ at the next moment as the input, then output the current activation h_t . The process can be formulated as

$$\overrightarrow{h}_t = \overrightarrow{\text{GRU}}(f_t, \overrightarrow{h}_{t-1}) \quad (1)$$

$$\overleftarrow{h}_t = \overleftarrow{\text{GRU}}(f_t, \overleftarrow{h}_{t+1}) \quad (2)$$

$$h_t = w_t \overrightarrow{h}_t + v_t \overleftarrow{h}_t + b_t. \quad (3)$$

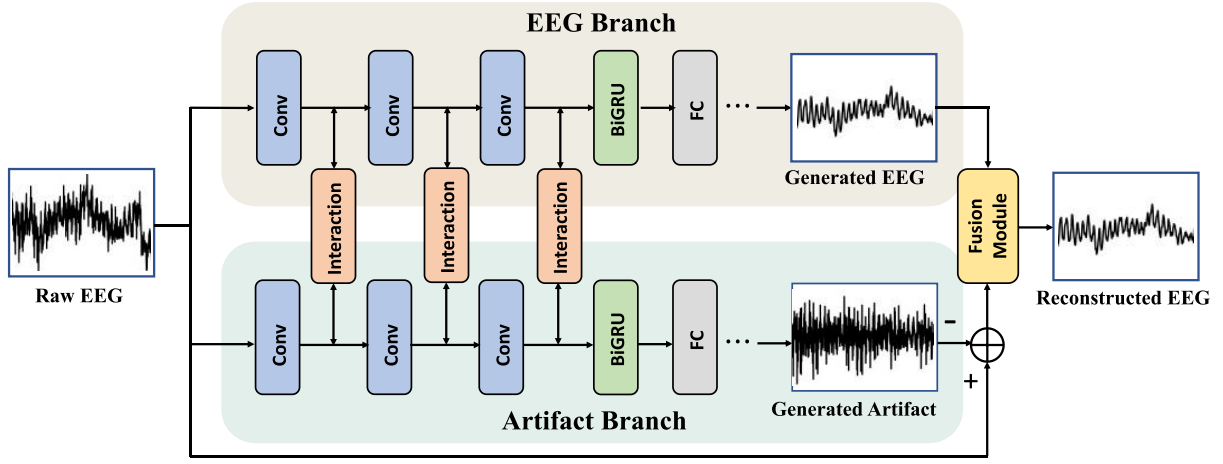


Fig. 1. Overall framework of the EEGIFNet.

By doing this, local feature $F_E = [f_1, f_2, \dots, f_d]$ is encoded to global feature $H_E = [h_1, h_2, \dots, h_d]$, which is further input into the fully connected layers to predict clean EEG. We assume that the contaminated EEG is a linear superposition of the clean EEG and artifacts. The two branches are given by

$$y = x + a \quad (4)$$

$$\tilde{x} = f_{\text{EEG}}(y) \quad (5)$$

$$\tilde{a} = f_{\text{Artifact}}(y) \quad (6)$$

where $x \in \mathbb{R}^{1 \times T}$ and $a \in \mathbb{R}^{1 \times T}$ are the clean EEG and artifact, respectively. $\tilde{x} \in \mathbb{R}^{1 \times T}$ and $\tilde{a} \in \mathbb{R}^{1 \times T}$ are the predicted EEG and artifact, respectively.

B. Interaction Module

Considering that the two branches share the same input, there are redundant artifact components and missing EEG components in the EEG branch inevitably, and vice versa. To learn complementary information of EEG and artifact, we design the interaction module for information communication in intermediate feature domains. Through this module, the features transformed from the other branches are expected to improve their own parts. In our work, interaction modules are added after each convolutional layer.

As shown in Fig. 2, there are two paths in the interaction module, namely EEG-to-artifact and artifact-to-EEG. Let $F_E^i \in \mathbb{R}^{C \times d_i}$ and $F_A^i \in \mathbb{R}^{C \times d_i}$ denote the EEG feature and artifact feature extracted from the i th convolutional layer of the two branches, where $i = 1$ to 3, C is the number of filters and d_i is the length of the feature. Taking artifact-to-EEG path as an example, the two features are concatenated in channel dimension and then input to a convolutional layer with C channels, followed by a Sigmoid activate function to learn a multiplication mask matrix $M_A^i \in \mathbb{R}^{C \times d_i}$ between 0 and 1, predicting the information that needs to be suppressed and retained of F_A^i . The sigmoid activation function can map any input to a value between 0 and 1. In the LSTM unit, a sigmoid is used to obtain a memory gate that can determine the retention ratio of the content at the previous moment. If the

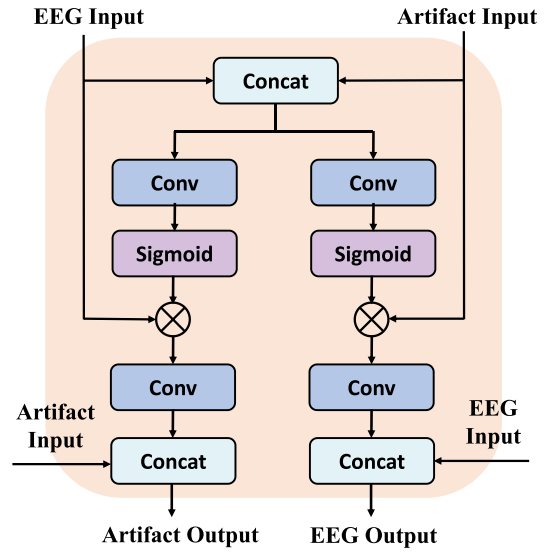


Fig. 2. Structure diagram of the interaction module.

value is 0, all the information of the corresponding location is forgotten, and if the value is 1, all the information of the corresponding location is retained. Inspired by this, we use the sigmoid activation function to generate the mask matrix. A preliminary residual feature can be obtained by multiplying F_A^i and M_A^i elementally. We use another convolutional layer with C' channels to compress the number of channels and get high-level residual representation $R_{A2E}^i \in \mathbb{R}^{C' \times d_i}$. Lastly, enhanced EEG feature $F_{E_{\text{new}}}^i \in \mathbb{R}^{(C+C') \times d_i}$ is generated by concatenating F_E^i and R_{A2E}^i . Here, C and C' are equal to 32 and 4, respectively. The process can be described as

$$M_A^i = \text{Sigmoid}(\text{Conv}(F_E^i || F_A^i)) \quad (7)$$

$$R_{A2E}^i = \text{Conv}(F_A^i * M_A^i) \quad (8)$$

$$F_{E_{\text{new}}}^i = F_E^i || R_{A2E}^i \quad (9)$$

where $(||)$ denotes concatenation operation, $(*)$ is element-wise multiplication. $F_{E_{\text{new}}}^i$ is used as the input of the next convolutional layer or the BiGRU layer for the EEG branch. Similarly, we can get enhanced artifact features based on the EEG-to-artifact path.

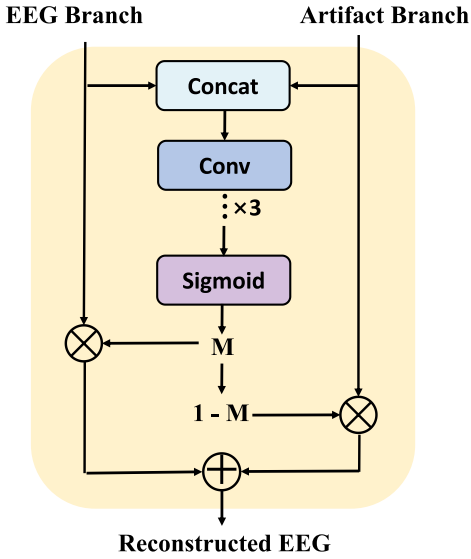


Fig. 3. Structure diagram of the fusion module.

C. Fusion Module

After each branch gets its output, we can obtain the predicted EEG $\tilde{x} \in \mathbb{R}^{1 \times T}$ and $\tilde{x}' \in \mathbb{R}^{1 \times T}$ in two ways. $\tilde{x}$ is directly generated by the EEG branch, mainly taking the EEG characteristics into account, while $\tilde{x}'$ is obtained by subtracting the generated artifact from the contaminated EEG, paying more attention to the artifact characteristics. We further propose a fusion module to adaptively combine the two outputs.

As shown in Fig. 3, $\tilde{x}$ and $\tilde{x}'$ are stacked and fed into three convolutional layers and a Sigmoid activation function to generate an element-wise mask $M \in \mathbb{R}^{1 \times T}$, which acts as a gate to control the retention ratio of the two outputs. The channel number of the three convolutional layers is set to 32, 32, and 1, respectively. Based on mask M , the reconstructed EEG $\hat{x} \in \mathbb{R}^{1 \times T}$ can be obtained. The process of the fusion module can be formulated as

$$\tilde{x}' = y - \tilde{a} \quad (10)$$

$$M = \text{Mask}(\tilde{x} || \tilde{x}') \quad (11)$$

$$\hat{x} = M * \tilde{x} + (1 - M) * \tilde{x}' \quad (12)$$

where $\text{Mask}()$ stands for convolution and Sigmoid operations.

D. Optimization of EEGIFNet

We employ three mean square error (mse) loss functions to optimize our model, which act on the EEG and artifact predicted by the two branches, and the EEG reconstructed by the fusion module, respectively. The overall loss function $\mathcal{L}(\theta_E, \theta_A, \theta_I, \theta_F)$ is defined as the following form:

$$\begin{aligned} \mathcal{L}(\theta_E, \theta_A, \theta_I, \theta_F) = & \text{mse}_F(\hat{x}, x; \theta_F) + \text{mse}_E(\tilde{x}, x; \theta_E, \theta_A, \theta_I) \\ & + \text{mse}_A(\tilde{a}, a; \theta_E, \theta_A, \theta_I) \end{aligned} \quad (13)$$

where $\theta_E, \theta_A, \theta_I$, and θ_F are parameters of the EEG branch, artifact branch, interaction module, and fusion module. mse_E and mse_A jointly optimize the first three parts, while mse_F

optimizes the fusion module. Taking mse_E as an example, the mean squared error loss is formulated as

$$\text{mse}_E(\tilde{x}, x; \theta_E, \theta_A, \theta_I) = \frac{1}{N} \sum_{i=1}^N \|\tilde{x}_i - x_i\|_2^2 \quad (14)$$

where N denotes the number of training samples.

III. EXPERIMENTS

A. Data Collection

The contaminated EEG is generated by a linear mixture of the clean EEG and real artifacts. The clean EEG signals, the EOG artifacts, and the EMG artifacts are obtained from the EEGdenoiseNet dataset, which contains 4514 EEG segments, 5598 EMG segments, and 3400 EOG segments [11]. All the data are single-channel with a duration of 2 s. Among them, the authors collected EEG data from a BCI dataset with 64-channel EEG and performed a series of preprocessing to get clean EEG, including filtering between 1 and 80 Hz, slicing, visual examination from experts, and so on. For the EOG artifacts, multiple publicly available EEG datasets with additional EOG channels are used. The EOG signals are bandpass filtered between 0.3 and 10 Hz. For the EMG artifacts, a facial EMG dataset is used since they are the main sources of myogenic artifacts. The raw EMG signals are bandpass filtered between 1 and 120 Hz and notched at the powerline frequency. The specific preprocessing of the EEG, EMG, and EOG can be referred to [11].

The ECG artifacts are obtained from the MIT-BIH arrhythmia dataset [18]. The dataset contains 48 half-hour excerpts of two-channel ECG recordings at a sampling frequency of 360 Hz, obtained from 47 subjects studied by the BIH Arrhythmia Laboratory. The ECG signals are filtered with a cut-off frequency of 45 Hz.

The motion artifacts are obtained from a public dataset contributed by Sweeney et al. [19]. The dataset consists of 24 two-channel EEG recordings from six subjects. The EEG signals are acquired by two closely placed electrodes. The motion artifacts are generated by pulling the cable of one electrode and filtered according to [20].

The PLI artifacts are obtained according to published work [21], which generates random PLI artifacts composed of the 50-Hz component and the harmonics sine functions at 100 and 150 Hz.

All these signals are resampled to 256 Hz and then divided into 1-D segments of 2 s. After that, we obtained 4514 EEG segments, 5598 EMG segments, 3400 EOG segments, 3600 ECG segments, 2760 motion segments, and 3600 PLI segments.

The contaminated EEG can be generated by

$$y = x + \lambda \cdot n \quad (15)$$

where n denotes real artifact and λ is used to control the SNR. SNR is defined as

$$\text{SNR} = 10 \log \frac{\|x\|_2^2}{\|\lambda \cdot n\|_2^2}. \quad (16)$$

By adjusting the coefficient λ , the SNR of the contaminated EEG ranges from -5 to 5 dB according to [12]. After that, we can obtain five types of contaminated EEG.

B. Evaluation Indicators

We select three commonly used indicators to evaluate the reconstructed EEG, including the correlation coefficient (CC), relative root mse (RRMSE), and SNR.

CC describes the linear correlation between the reconstructed EEG and clean EEG, with values ranging from -1 to 1 . A value closer to 1 corresponds to better performance. CC is defined as

$$CC = \frac{\text{Cov}(x, \hat{x})}{\sqrt{\text{Var}(x)\text{Var}(\hat{x})}} \quad (17)$$

where Cov and Var are the covariance and variance.

RRMSE is usually employed to evaluate the deviation between two signals. The lower value indicates the better performance. RRMSE can be expressed as

$$RRMSE = \frac{\|\hat{x} - x\|_2}{\|x\|_2}. \quad (18)$$

The SNR can be computed as

$$SNR = 10 \log \frac{\|x\|_2^2}{\|\hat{x} - x\|_2^2}. \quad (19)$$

The larger value of the SNR indicates better reconstruction quality.

C. Experimental Setup

After preprocessing, we can obtain clean EEG signals and five types of artifacts. We apply a tenfold cross-validation technique to evaluate the proposed method. All the data is randomly divided into ten parts. We use one part for the test and the rest for training. Taking EMG artifacts as an example, we divided 4514 EEG segments and 5598 EMG segments into training data and test data, respectively. In the training data, we randomly reused some EEG segments to match the number of EEG segments with EMG segments. The same operation is performed on the test data. Other artifacts are mixed similarly. 90% of the training data is used for the training process, and the remaining 10% is used as the validation set to select the optimal parameters. The average performance of the tenfold cross-validation is taken as the final result. To improve the diversity of data, EEG signals are randomly combined with each kind of artifact ten times at different SNR levels. For the training set, they are combined at a uniformly distributed SNR from -5 to 5 dB. For the validation set and the testing set, they are combined at -5 to 5 dB SNR with a step size of 1 dB. In this way, five kinds of contaminated EEGs are generated in our experiments. For each kind of EEG, we can obtain the paired data $(x, \lambda \cdot n, y)$, which can also be denoted as (x, a, y) , to train and test the proposed EEGIFNet end-to-end.

D. Implementation Details

All the convolutional layers in our model are followed by a batch normalization (BN) layer and a rectified linear unit

(ReLU) activation function. The kernel size for all the convolutional layers is 9 . In the two branches, all three convolution layers have 32 channels. The BiGRU layer has 32 hidden neurons in each direction. The number of neurons in each fully connected layer is 512 , which is equal to the length of EEG segments. To prevent overfitting, dropout with a rate of 0.1 is employed after each convolution and fully connected layer.

During training, we adopted the Adam optimizer to minimize the loss function. The epoch is 80 and the batch size is 256 . The learning rate is set to 5×10^{-5} according to [11] and [22]. Our model is implemented in Python 3.7 with PyTorch library, running on a Nvidia 3080 GPU. The code is available at <https://github.com/Heng-Cui/EEGIFNet>.

IV. RESULTS AND DISCUSSIONS

A. Quantitative Results

The proposed EEGIFNet model is compared with several state-of-the-art DL methods under the same experimental setting, including the FCNN [10], simple CNN (SCNN) [11], 1-D residual convolutional neural network (1D-ResCNN) [12], RNN [11], and novel convolutional neural network (Novel CNN) [22]. Previous studies have shown that DL methods significantly outperform traditional methods, such as ICA, WT, EEMD-ICA, and EEMD-CCA [12], [14], [23]. Therefore, traditional methods are not included in our experiments. We report the experimental results of five tasks, including EOG artifact removal, EMG artifact removal, ECG artifact removal, motion artifact removal, and PLI artifact removal.

Table I presents the average CC, RRMSE, and SNR on five tasks at all SNR levels. For EMG artifact removal, the FCNN, SCNN, 1D-ResCNN, and RNN achieve similar performance. Novel CNN and EEGIFNet are obviously superior to them in all indicators. Since there are numerous sources in EMG artifacts and their frequency distribution largely overlaps with that of EEG, EMG artifact removal is extremely challenging. Novel CNN uses a deep network structure with 14 convolution layers for continuous artifact reduction. On the contrary, our model obtains better performance based on the dual-branch structure. Furthermore, we compared the performance of different methods at various SNR levels. The results are shown in Fig. 4(a). We can find that the performance of all methods is improved with the increase of the SNR. Although the average results of the first four methods are similar, FCNN and RNN are more suitable for low SNR levels, while SCNN and 1D-ResCNN are more prominent at high SNR levels. Compared with Novel CNN, our model performs better when the SNR is greater than 0 dB.

As shown in Table I, for EOG artifact removal, FCNN and RNN perform worse than CNN-based models, possibly because the CNN is better at capturing local details. Different from EMG artifacts, the complex Novel CNN has no advantage in removing simpler EOG artifacts. EEGIFNet obtains the highest CC, the lowest RRMSE, and the highest SNR. The CC, RRMSE, and SNR of the EOG artifact removal experiment at different SNR levels are shown in Fig. 4(b). Our model remarkably outperforms other methods at all SNR

TABLE I
AVERAGE CC, RRMSE, AND SNR (dB) OF THE VARIOUS METHODS ON FIVE ARTIFACT REMOVAL TASKS

Artifact	Metrics	FCNN [10]	SCNN [11]	1D-ResCNN [12]	RNN [11]	Novel CNN [22]	EEGIFNet
EMG	CC	0.872	0.886	0.868	0.873	0.918	0.926
	RRMSE	0.480	0.467	0.489	0.488	0.364	0.356
	SNR	6.828	7.082	6.628	6.736	9.227	9.463
EOG	CC	0.864	0.939	0.936	0.849	0.917	0.974
	RRMSE	0.496	0.307	0.309	0.520	0.357	0.210
	SNR	6.571	10.951	11.285	6.024	10.112	14.653
ECG	CC	0.826	0.923	0.924	0.824	0.871	0.951
	RRMSE	0.555	0.360	0.360	0.557	0.425	0.273
	SNR	5.596	9.538	9.502	5.429	8.548	12.567
motion	CC	0.861	0.922	0.921	0.848	0.883	0.945
	RRMSE	0.507	0.376	0.370	0.522	0.427	0.286
	SNR	6.343	9.116	9.451	5.808	8.464	11.951
PLI	CC	0.915	0.985	0.983	0.926	0.871	0.994
	RRMSE	0.416	0.160	0.161	0.402	0.381	0.106
	SNR	8.133	16.286	16.340	8.397	10.573	20.024

levels. Notably, EEGIFNet achieves a CC value of 0.963, an RRMSE value of 0.250, and an SNR value of 12.80 at -5 dB SNR level. This result even exceeds the performance of other models at the 5-dB SNR level, indicating that EEGIFNet works robustly in the face of strong EOG artifacts.

Similarly, for ECG and motion artifact removal, FCNN and RNN perform poorly, followed by Novel CNN. Although SCNN and 1D-ResCNN achieve better average results on all indicators, they are not dominant at low SNR levels as shown in Fig. 4(c) and (d). Conversely, the performance of EEGIFNet is optimal at all SNR levels.

For PLI artifact removal, Novel CNN obtains a CC value of 0.871, which is the only one less than 0.9 among all methods. The results prove that the large-scale network with numerous parameters is redundant for dealing with simple artifacts. EEGIFNet achieves the best performance again, with CC higher than 0.99, RRMES lower than 0.11, and SNR higher than 20 dB. As shown in Fig. 4(e), when the SNR is greater than 0 dB, the performance of SCNN and 1D-ResCNN is no longer improved with the increase of SNR level, or even decreased. Compared with them, EEGIFNet has obvious advantages at high SNR levels.

B. Ablation Experiments

Quantitative results indicate that EEGIFNet outperforms the compared methods on diverse artifacts and SNR levels. To verify the effectiveness of each component, we conducted additional ablation experiments on five artifact removal tasks. Three simplified models are compared by discarding several important parts. They are defined as follows: 1) ENet, which uses the EEG branch of EEGIFNet to directly predict pure EEG; 2) ANet, which uses the artifact branch of EEGIFNet to predict artifacts and obtains pure EEG by subtracting predicted artifacts from contaminated EEG; and 3) FNet, which is a two-branch fusion model without the interaction module.

Table II shows the average CC, RRMSE, and SNR on five artifact removal tasks at various SNR levels. Comparing the two single-branch networks, it can be found that ENet

TABLE II
AVERAGE CC, RRMSE, AND SNR (dB) OF THE ABLATION EXPERIMENTS ON FIVE ARTIFACT REMOVAL TASKS

Artifact	Metrics	ENet	ANet	FNet	EEGIFNet
EMG	CC	0.910	0.829	0.920	0.926
	RRMSE	0.393	0.611	0.375	0.356
	SNR	8.894	5.183	9.090	9.463
EOG	CC	0.928	0.966	0.966	0.974
	RRMSE	0.332	0.251	0.217	0.210
	SNR	10.158	12.561	14.186	14.653
ECG	CC	0.925	0.903	0.943	0.951
	RRMSE	0.344	0.432	0.282	0.273
	SNR	10.177	8.094	12.042	12.567
motion	CC	0.921	0.923	0.941	0.945
	RRMSE	0.352	0.339	0.291	0.286
	SNR	10.045	10.549	11.279	11.951
PLI	CC	0.959	0.991	0.991	0.994
	RRMSE	0.236	0.124	0.106	0.106
	SNR	12.583	18.468	19.977	20.024

performs better in the EMG and ECG artifact removal tasks, while ANet is better at removing EOG, motion, and PLI artifacts. The results show that for different artifacts, the contributions of the two branches are different. We can also see that FNet is superior to each single-branch model on five tasks, indicating the effectiveness of fusing the outputs of the two branches. By comparing FNet with EEGIFNet, we can observe varying degrees of improvement in almost all indicators after introducing the interaction module. These results verify that the information exchange between the two branches is beneficial to both sides.

C. Visualizations

Furthermore, to intuitively verify the effectiveness of the two-branch fusion model, we visualize the EEG predicted by the EEG branch, artifact branch, and fusion module of EEGIFNet on five artifact removal tasks. The results are shown in Fig. 5. Here, the output of the artifact branch

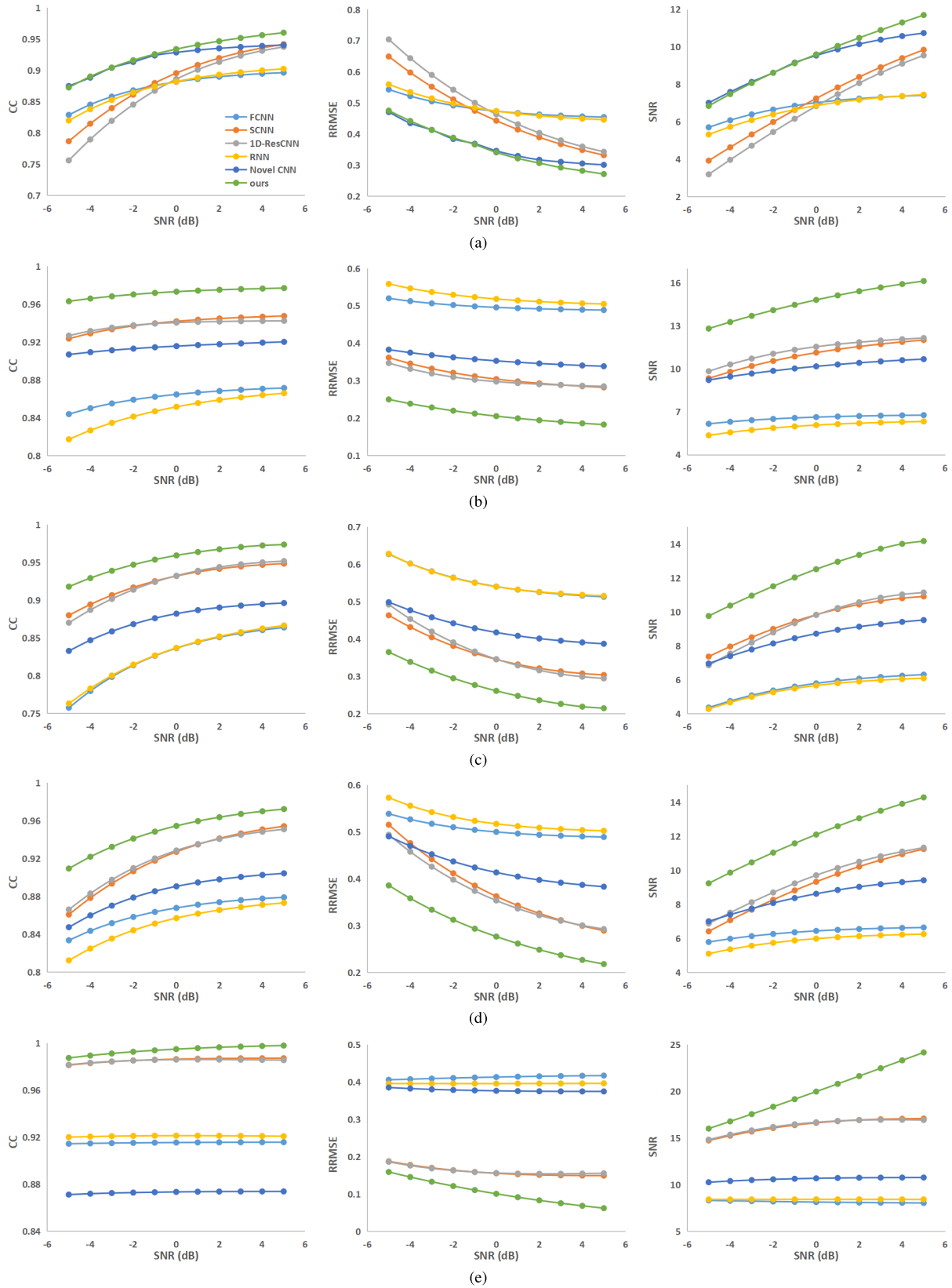


Fig. 4. Performance comparison at various SNR levels in terms of CC, RRMSE, and SNR on (a) EMG, (b) EOG, (c) ECG, (d) motion, and (e) PLI artifact removal tasks. The horizontal axis represents the SNR with unit dB.

can be obtained by subtracting the predicted artifact from the contaminated EEG. For EMG artifact removal, we can observe from Fig. 5(a) that there is EEG distortion in the output of the EEG branch when the artifact is weak (0–200 sampling points), and there is artifact residue in the output of the artifact branch when the artifact is strong (200–512



Fig. 5. Denoised EEG (orange) compared with the original EEG (blue) and clean EEG (green) of the 1) EEG branch, 2) artifact branch, and 3) fusion module. (a) EMG. (b) EOG. (c) ECG. (d) Motion. (e) PLI.

sampling points). After fusing the outputs of the two branches, we can not only retain more EEG information, but also remove residual artifacts as much as possible. Similar situations occur in EOG, ECG, and motion artifact removal tasks. For PLI artifact removal, compared to the EEG branch, the output of EEGIFNet (orange) has more overlapping regions with the clean EEG (green), indicating the effectiveness of the dual-branch strategy.

D. Predict EEG or Artifact?

Benefiting from the simultaneous predictions of EEG and artifacts, the proposed EEGIFNet achieves optimal performance in all indicators. The results of ablation experiments and visualizations show that the EEG branch and artifact branch have their own advantages. In this section, we will investigate whether it is better for EEG denoising to predict EEG or artifacts. EEG prediction model ENet and artifact

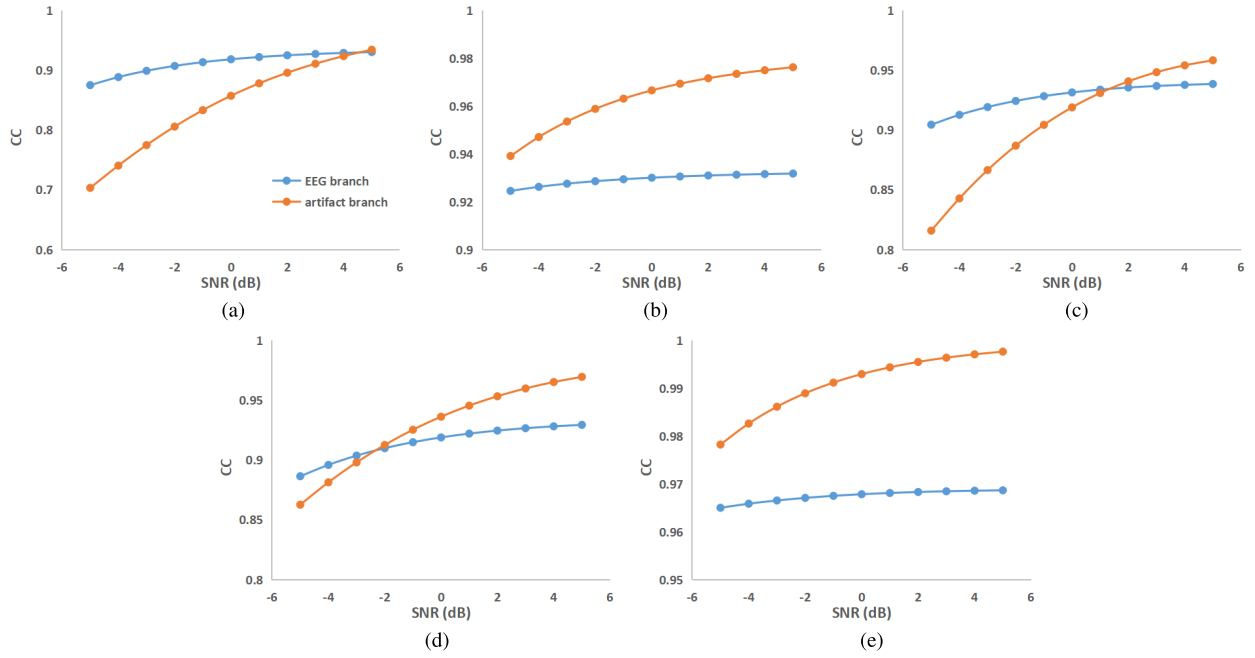


Fig. 6. Performance comparison at various SNR levels between EEG prediction models and artifact prediction models by using FCNN and 1D-ResCNN on (a) EMG, (b) EOG, (c) ECG, (d) motion, and (e) PLI artifact removal tasks. The horizontal axis represents the SNR with unit dB.

TABLE III

AVERAGE CC OF ENET AND ANET ON FIVE KINDS OF ARTIFACTS AND THE PAIRED T-TEST STATISTICS ANALYSIS BETWEEN THEM AT THE SIGNIFICANCE LEVEL OF 0.05. (*) REPRESENTS A SIGNIFICANT DIFFERENCE BETWEEN THE TWO METHODS

Artifact	CC		p_value
	ENet	ANet	
EMG	0.910*	0.829	2.78e-03
EOG	0.928	0.966*	5.17e-07
ECG	0.925	0.906	6.44e-02
motion	0.921	0.923	8.19e-02
PLI	0.959	0.991*	3.28e-08

prediction model ANet are selected in our experiments. The EEG predicted by ANet is obtained by subtraction strategy. For simplification, we only show the average CC on five tasks. From Fig. 6, it can be found that for EMG artifact removal, the performance of ANet is significantly lower than that of ENet at low SNR levels, and tends to approach with the improvement of SNR level. For EOG and PLI artifact removal, ANet outperforms ENet, and the gaps between them increase with the improvement of SNR level. For ECG artifact removal, ENet performs better at low SNR levels, while ANet performs better at high SNR levels. They achieve similar performance at an SNR of -1 dB. For motion artifact removal, ANet becomes superior to ENet when the SNR level is greater than -2 dB. We can roughly observe two rules. One is that when faced with complex artifacts, it is more suitable to directly predict EEG, otherwise predicting artifacts is a smarter choice. The other is that it is better to predict the EEG at low SNR levels, while it is better to predict artifacts at high SNR levels.

We further perform paired t-test statistical analyses of our findings. As shown in Table III, ENet is significantly superior to ANet in removing EMG artifacts, while ANet has significant

TABLE IV

AVERAGE CC OF ENET AND ANET AT DIFFERENT SNR LEVELS ON ALL ARTIFACTS AND THE PAIRED T-TEST STATISTICS ANALYSIS BETWEEN THEM AT THE SIGNIFICANCE LEVEL OF 0.05. (*) REPRESENTS A SIGNIFICANT DIFFERENCE BETWEEN THE TWO METHODS

SNR levels	CC		p_value
	ENet	ANet	
< 0 dB	0.922*	0.894	2.71e-02
> 0 dB	0.938	0.957*	5.23e-04

advantages in removing EOG and PLI artifacts. For ECG and motion artifacts, there is no significant difference between the two methods. These results are consistent with the first rule. Table IV shows that when the SNR of the added noise is less than 0 dB, the improvement of ENet over ANet is statistically significant. When the SNR is greater than 0 dB, ANet has significant advantages. The results are consistent with the second rule. These findings can provide a novel perspective and foundational reference for DL work of EEG denoising in the future. A reasonable explanation is that to the extreme, predicting a clean EEG from raw signals with simple artifacts or high SNR is close to an identity mapping. It is much easier to learn a residual approaching zero than to fit an identity mapping with a bunch of nonlinear layers [24]. This can also explain why the artifact branch works prominently in areas with weak artifacts in visualizations, such as 0–200 sampling points in Fig. 5(a), 0–150 sampling points in Fig. 5(b) and (c).

E. Generalization of EEGIFNet

To verify the generalization ability of the proposed model, we conduct additional cross-dataset experiments, which ensure that training and test data come from different subjects and

TABLE V
AVERAGE CC, RRMSE, AND SNR (dB) OF THE VARIOUS METHODS ON CROSS-DATASET EXPERIMENT

Artifact	Metrics	FCNN [10]	SCNN [11]	1D-ResCNN [12]	RNN [11]	Novel CNN [22]	EEGIFNet
EMG	CC	0.656	0.770	0.770	0.696	0.614	0.819
	RRMSE	0.760	0.613	0.614	0.718	0.781	0.586
	SNR	2.479	4.507	4.451	3.008	2.226	4.714
EOG	CC	0.628	0.736	0.709	0.644	0.589	0.784
	RRMSE	0.773	0.649	0.682	0.757	0.792	0.588
	SNR	2.324	3.982	3.486	2.570	2.090	4.949

TABLE VI
PARAMETER QUANTITIES, FLOPS, AND COMPUTATION TIME OF DIFFERENT MODELS

	FLOPS ($\times 10^6$)	Parameters ($\times 10^6$)	Time (ms)
FCNN	1.049	1.051	0.018
SCNN	36.274	16.816	0.793
1D-ResCNN	59.523	25.231	1.369
RNN	0.799	0.788	0.156
Novle CNN	307.331	33.560	2.260
EEGIFNet	100.784	5.888	1.819

TABLE VII
AVERAGE CC, RRMSE, AND SNR (dB) OF THE VARIOUS METHODS FOR SIMULTANEOUS REMOVAL OF MULTIPLE ARTIFACTS

	CC	RRMSE	SNR
FCNN	0.837	0.542	5.711
SCNN	0.866	0.494	6.491
1D-ResCNN	0.846	0.535	5.853
RNN	0.832	0.553	5.409
Novle CNN	0.879	0.431	8.130
EEGIFNet	0.902	0.416	8.093

devices. All models are trained on the EEGdenoiseNet dataset. For the test data, the EEG and EOG are obtained from the Semisimulated EEG/EOG Dataset [25], which contains EEG, vertical EOG, and horizontal EOG recorded from 27 subjects. The EEG is bandpass filtered between 0.5 and 40 Hz, notched at powerline frequency. EOG is bandpass filtered between 0.5 and 5 Hz. The EMG data is obtained from [4]. It collected EMG signals at a sampling frequency of 512 Hz from the forearms of 23 healthy subjects. The EMG is bandpass filtered between 1 and 120 Hz and notched at the powerline frequency. All data are resampled to 256 Hz and cut into single-channel 2-s segments. The EEG and artifacts are mixed at -5 to 5 dB SNR to generate contaminated EEG.

The experimental results are shown in Table V. Due to the impact of individuals, equipment, and environment, all indicators have declined to varying degrees. Each method performs better at removing EMG than at removing EOG, which may be due to greater variation in EOG across subjects. Novel CNN performs worst on both artifacts since it overfits the training data. EEGIFNet is superior to other methods on all indicators. The CCs of EEGIFNet (0.819 on EMG and 0.784 on EOG) are significantly better than those of the second-ranked SCNN (0.770 on EMG and 0.736 on EOG).

F. Computation Efficiency

We compare the parameter quantities, floating-point operations per second (FLOPS), and computation time for a single implementation of different models. The implementation is done in Python and runs under 64-bit Microsoft Windows 10 OS on a computer with Intel¹ Core² i5-9400 2.90 GHz CPU and 16.00 GB RAM. As can be seen from Table VI, EEGIFNet has 5.888 M parameters, only higher than FCNN and RNN. The determinant of the parameter quantities is the

number of neurons in the last fully connected layers. SCNN, 1D-ResCNN, and Novel CNN contain a large number of neurons, which can easily lead to overfitting and affect the generalization ability of the model. The single implementation time of EEGIFNet is 1.819 ms. Although this result is higher than the FCNN, SCNN, 1D-ResCNN, and RNN, it is far less than the duration of input data (2 s), indicating that our model can process EEG data online. Besides, the FLOPS of EEGIFNet is 100.784 M. The classic MobileNet network deployed on the mobile has a FLOPS of 569 M [26]. This means that EEGIFNet has the potential to be deployed in wearable EEG devices. Overall, the proposed method contributes to enhancing the reliability of the EEG acquisition instrument system and improving the accuracy of subsequent signal analysis in the I&M field.

G. Remove Multiple Artifacts Simultaneously

We discuss the robustness of the proposed model for the simultaneous removal of multiple artifacts. We mixed five artifacts simultaneously into clean EEG in the range from -5 to 5 dB SNR. The experimental results are shown in Table VII. Novel CNN performs better than other comparison methods, indicating that deeper models are suitable for solving complex artifacts. By fitting EEG and artifacts with two branches, EEGIFNet achieves almost optimal performance. It is reasonable to believe that designing a multibranch structure to model each artifact separately is a feasible scheme to remove multiple artifacts simultaneously in a single step. Nevertheless, this needs further comprehensive verification in the future.

V. CONCLUSION

In this article, a novel dual-branch interactive fusion neural network, called EEGIFNet is proposed for EEG denoising. Clean EEG and residual artifacts are predicted by the two branches, respectively. An interaction module is proposed for

¹Registered trademark.

²Trademarked.

information exchange and feature enhancement. A fusion module is introduced for adaptively fusing the outputs of the two branches. We conduct extensive experiments on EMG, EOG, ECG, PLI, and motion artifact removal tasks. The results show that the proposed EEGIFNet outperforms the state-of-the-art methods on diverse artifacts and SNR levels, including FCNN, SCNN, 1D-ResNet, RNN, and Novel CNN. In addition, based on ablation experiments and visualizations, we observe that the EEG branch and artifact branch have their own advantages for different artifacts and periods. Furthermore, we investigate how to choose between EEG prediction and artifact prediction. Experiments show that predicting clean EEG is more effective when faced with signals with complex artifacts or low SNR, otherwise predicting artifacts. This finding can provide a different perspective and benchmark for future DL work on EEG denoising. In practice, the collected EEG is often accompanied by several artifacts. In the future, we will improve the proposed model to handle multiple types of artifacts simultaneously.

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